

Fatih Berkay Sarpkaya

Education

- 2022–Present **Doctor of Philosophy (Ph.D.)**, *New York University Tandon School of Engineering*, Brooklyn, NY
Expected Graduation: May 2027
Major: Electrical & Computer Engineering | Advisor: [Professor Shivendra Panwar](#) | **GPA: 3.949/4.00**
- 2017–2022 **Bachelor of Science (B.Sc.)**, *Bilkent University*, Ankara, Turkey
Major: Electrical and Electronics Engineering, **CGPA: 3.77/4.00**

Research Interests

Transport Protocols & Congestion Control | Low-Latency Networking (L4S) | Internet Architecture & Traffic Engineering | Data Center & AI Networking | Network Performance Modeling & Measurement

Work & Research Experience

- Summer 2026 **Microsoft Research**, *AI Network Observability Research Intern*, Redmond, WA
Performance Modeling and Analysis of AI Accelerator Infrastructure — Managers: Anqi Guo, Nicholas Lurski, Pratik Marolia
- Developed and calibrated an FPGA-based emulation framework for AI accelerator collective communication workloads.
 - Modeled communication pipelines and dependencies for collective operations, validating emulation behavior against hardware measurements.
 - Characterized collective communication performance across message sizes, identifying synchronization and communication overheads affecting end-to-end latency.
- Summer 2025 **Nokia Bell Labs**, *Network Reliability Research Intern (Summer) / Co-op (Fall)*, Murray Hill, NJ
- Fall 2025 **Congestion Control for Public High-Performance WANs** — Managers: Andrea Francini, Bilgehan Erman, Gary Atkinson
- Evaluated CUBIC, BBRv1, and BBRv3 for predictable large data transfers over public high-performance WANs. [\[HPSR'26\]](#)
 - Designed large-scale FABRIC testbed experiments using Linux and DPDK-based traffic shaping and measurement tools; modified and optimized Linux kernel congestion control modules.
- Summer 2024 **InterDigital**, *Research & Innovation Wireless Intern*, New York, NY
Flexible Packet Coding Performance Analysis & Generative Semantic Communication Literature Review — Manager: Pascal Adjakple
- Implemented various network coding and packet duplication simulations in MATLAB, and analyzed performance results.
 - Conducted a literature review on generative AI in semantic communication and its potential impact on modern communication systems.
- Sep. 2022 – Present **NYU Wireless Research Center**, *Graduate Researcher*, Brooklyn, NY
- Transport & Low-Latency Networking: evaluation and modeling of congestion control coexistence across BBR, TCP Prague, CUBIC/Reno, AQMs, and partial L4S deployments [\[IFIP Networking'25\]](#) [\[PAM'25\]](#) [\[ANRW'24\]](#)
 - Internet Architecture & Traffic Engineering: multi-domain low-latency service using traffic isolation, SRv6 path steering, and BGP signaling [\[ICNP'26\]](#)
 - Reliable Communications: adversarial channel capacity and robust communication using random interleaving and thresholded LLR decoding [\[Asilomar'24\]](#)

Publications

- Oct. 2026 **Extending Low Latency Service Across the Internet**
Harkirat Singh*, **Fatih Berkay Sarpkaya***, Hakan Gulec, Fraida Fund, Shivendra Panwar
IEEE International Conference on Network Protocols (ICNP 2026) *Equal contribution [\[Link\]](#)

- June 2026 **Inherited Fragility in Dual-Queue AQMs under Heterogeneous Congestion Control**
Fatih Berkay Sarpkaya, Fraida Fund, Shivendra Panwar
Under Review
- June 2026 **Evaluation of TCP Congestion Control for Public High-Performance Wide-Area Networks**
Fatih Berkay Sarpkaya, Andrea Francini, Bilgehan Erman, Shivendra Panwar
IEEE High Performance Switching and Routing (HPSR 2026) [\[Link\]](#)
- May 2025 **BBR's Sharing Behavior with CUBIC and Reno**
Fatih Berkay Sarpkaya, A. Srivastava, Fraida Fund, Shivendra Panwar
IFIP Networking 2025 [\[Link\]](#)
- March 2025 **To Adopt or Not to Adopt L4S-Compatible Congestion Control? Understanding Performance in a Partial L4S Deployment**
Fatih Berkay Sarpkaya, Fraida Fund, Shivendra Panwar
Passive and Active Measurement Conference (PAM 2025) [\[Link\]](#)
- Oct. 2024 **Capacity of a Binary Channel with a Time-Bounded Adversary**
M. Ying, **Fatih Berkay Sarpkaya**, S. Bakirtas, Elza Erkip, T. Rappaport, Sundeep Rangan
58th Asilomar Conference on Signals, Systems, and Computers [\[Link\]](#)
- July 2024 **To Switch or Not to Switch to TCP Prague? Incentives for Adoption in a Partial L4S Deployment**
Fatih Berkay Sarpkaya, A. Srivastava, Fraida Fund, Shivendra Panwar
Applied Networking Research Workshop (ANRW 2024) [\[Link\]](#)

Relevant Coursework and Tools

- NYU - ECE Probability & Stochastic Processes | Internet Architecture & Protocols | Network Design and Algorithms | Data Center & Cloud Computing | Machine Learning | Network Modeling and Analysis | Digital Communication | Wireless Communication | Introduction to Wireless Networking | AI/ML for Network Traffic Analysis
- Bilkent - EE Computer Networks | Telecommunications I | Telecommunications II | Probability and Statistics | Statistical Learning and Data Analytics
- Tools Python | MATLAB | Linux | FABRIC Testbed | DPDK | CloudLab | GOAL

Awards & Honors

- 2026 **Dante Youla Award for Graduate Research Excellence**, NYU Electrical and Computer Engineering
- 2022-Present School of Engineering **Fellowship** at NYU Tandon School of Engineering
- 2017-2022 Comprehensive **Scholarship** at Bilkent University
- 2017 Ranked **450th** in **University Entrance Exam (LYS)** among 2 million high school students.

Teaching & Mentoring

- 2023–2024 **Course Assistant**, *Internet Architecture & Protocols; Communication Networks*, NYU Tandon School of Engineering
- Summer 2023 **Research Mentor**, *ARISE High School Summer Research Program*, NYU Tandon School of Engineering

Additional Activities

- 2018-2022 Active Member of Bilkent University IEEE Student Branch
- July 2019 Engaged in short-term volunteer work supporting children with disabilities through the European Voluntary Service (EVS) in Macedonia.